

Assessment: End-to-End Enterprise AI Application

Scenario: Intelligent Invoice Processing Assistant

A finance operations team receives hundreds of invoices daily through email and manual uploads. Processing these invoices manually is time-consuming and error-prone.

Your task is to build an AI-powered enterprise application that:

- extracts invoice information using OCR
 - processes the extracted content using an LLM
 - generates business-friendly summaries and insights
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Problem Statement

Build an end-to-end AI application where users can upload invoice documents through a frontend application.

The application should:

1. Upload invoice/image/PDF
2. Send document to Azure OCR service
3. Extract invoice fields
4. Pass OCR response to Azure OpenAI
5. Generate:
 - invoice summary
 - important business details
 - anomaly/warning detection
6. Display final output in frontend

Expected Architecture

```
React Frontend
↓
.NET Backend API
↓
Azure Document Intelligence (OCR)
↓
Extracted JSON/Text
↓
Azure OpenAI
↓
Business Summary / Insights
↓
Frontend Display
```

Functional Requirements

Part 1 — Frontend

Build a simple React UI where user can:

- upload invoice PDF/image
- click “Process Invoice”
- view:
 - extracted fields
 - AI-generated summary
 - warnings/anomalies

Part 2 — Backend (.NET)

Backend should:

- receive uploaded file
- call Azure OCR API
- extract:
 - vendor name
 - invoice number
 - invoice date
 - total amount

Then:

- send extracted content to Azure OpenAI

Part 3 — LLM Processing

Use prompt engineering to:

- summarize invoice
- identify suspicious values
- generate business insights

Example Prompt

```
You are a finance assistant.  
  
Analyze the following invoice data and provide:  
1. Invoice summary  
2. Important financial details  
3. Any suspicious or missing information  
  
Return response in structured bullet points.  
  
Invoice Data:  
[OCR OUTPUT]
```

Expected Output Example

```
Invoice Summary:  
- Vendor: ABC Pvt Ltd  
- Invoice Amount: ₹15,000  
- Invoice Date: 10-May-2026  
  
Warnings:  
- Tax field missing  
- High invoice amount compared to normal threshold  
  
Business Insight:  
- Payment approval may require manager review
```

Technical Requirements

Students should demonstrate:

- API consumption
- OCR integration
- LLM prompting
- Backend orchestration
- JSON handling
- Business logic filtering

Optional Enhancements

Students can additionally implement:

- Blob Storage upload
- Downloadable summary report
- Speech-based invoice narration
- Chat interface for invoice Q&A

Evaluation Criteria

Criteria	Marks
Frontend Functionality	20
Backend Integration	25
OCR Integration	20
Prompt Engineering	15
Response Formatting	10
Architecture Explanation	10

React Boilerplate code

JavaScript

```
      type="file"
      onChange={(e) => setFile(e.target.files[0])}
    />

    <br /><br />

    <button onClick={handleUpload}>
      Process Invoice
    </button>

    <br /><br />

    <pre>{result}</pre>

  </div>
);
}

export default App;
```